



HEALTH SAFETY & ENVIRONMENTAL

OCCIDENTAL OIL & GAS (OOG)

Guideline for Consequence Development in PHA Studies

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1. PURPOSE

The purpose of this guideline is to promote consistency in the development of “Consequences” which are postulated to result from credible Event Scenarios identified in typical Process Hazard Analysis (PHA) studies.

2. APPLICABILITY

This guideline applies to PHAs and other studies that involve development of Event Scenarios for Consequence analysis or risk assessment that are conducted within Occidental Oil & Gas (OOG) over which OOG has operational control.

3. CONSEQUENCE & EVENT SCENARIO RELATIONSHIP

Development of Event Scenarios is a critical first step in evaluating Risk. An Event Scenario is not necessarily the worst-case scenario but rather a realistic event that has a reasonable likelihood to result in harm to people, the environment or Company assets or earnings.

Generally, an Event Scenario is expressed as a Cause-Consequence combination that could lead to an undesirable outcome. The OOG Guideline 60.400.301.03 – Development of Event Scenarios separately provides guidance on the recommended steps that should be taken to develop an Event Scenario, with emphasis on general credibility of causes and types of Consequences to analyze for process and non-process systems.

Assessing the Consequences appropriately is key for an effective PHA. Underestimating Consequences can lead to insufficient risk reduction measures being applied and risk being insufficiently managed. Overestimating Consequences can lead to more risk reduction measures being applied than are warranted for the risk level, resulting in additional cost, inspection, testing, and maintenance requirements over the lifecycle of the operation.

The focus of this guideline is to elaborate on Consequence development for Event Scenarios with the goal to promote consistency in the Consequence Level assignment for various commonly analyzed scenarios estimated during PHA studies using the OOG Risk Matrix.

It should be noted that while some conventions listed in this guideline may be readily applied during a PHA session, others, such as those involving hole size assumptions and dispersion modeling, may require some pre-work or could be applied after the PHA study as part of refining Consequence estimates.



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4. CONSEQUENCE DEVELOPMENT GUIDELINES

4.1 Process Deviations

Figure 1: Operating Ranges and Limits illustrates how a typical process is controlled within defined safe upper and lower limits via a control system and human actions.

Theoretically, a process that is designed, constructed, operated, and maintained correctly should not deviate beyond safe upper or lower limits. In reality, process deviations that breach these limits may occur due to equipment failure (including mechanical, electrical and instrumentation), human error or external events. If a process deviates from safe upper and / or lower limits, there is a potential for loss of primary containment (LOPC) or release of energy that may result in adverse Consequences.

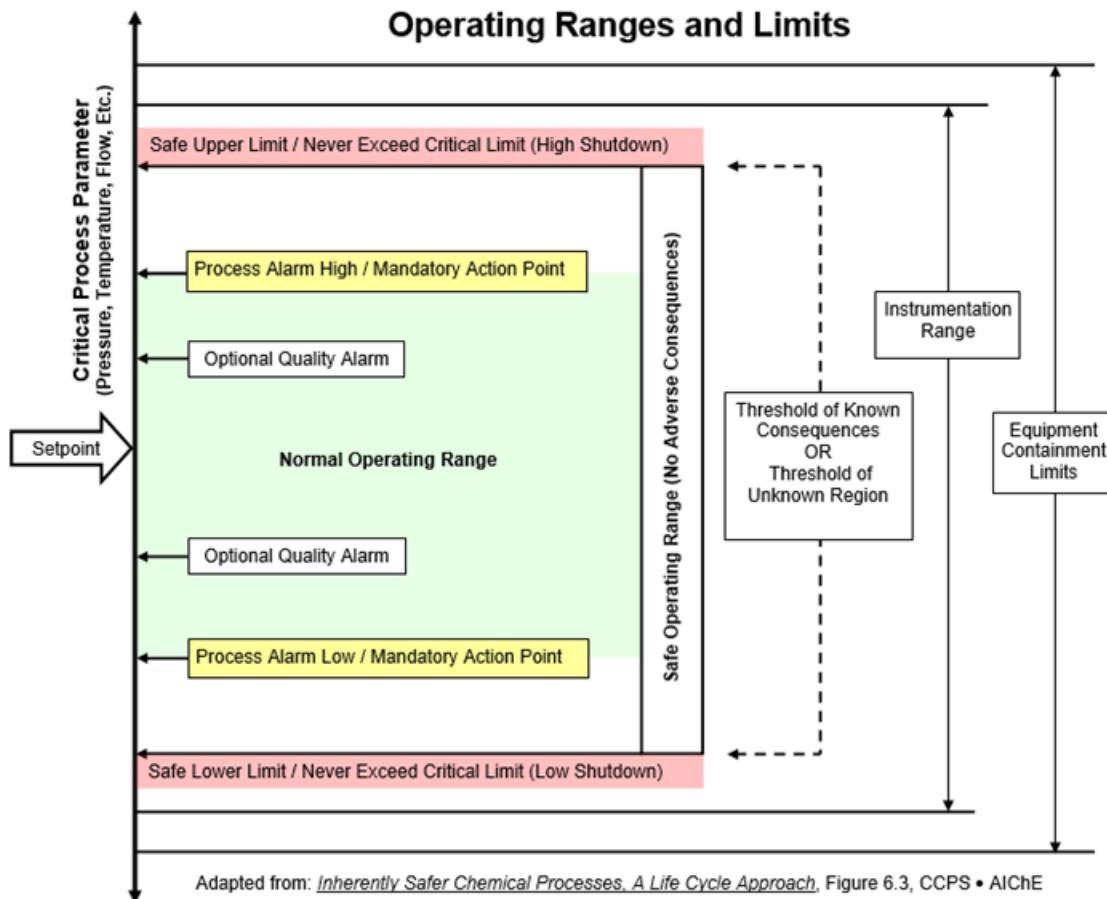


Figure 1: Operating Ranges and Limits

4.2 Consequence Analysis in PHAs & Possible Knock-on Effects

Figure 2: Example of LOPC Consequence Analysis illustrates how a process deviation could cause a vessel rupture and LOPC. In this simplified example, the analysis identifies four distinct types of Event Scenarios, each of which could lead to one or more Consequences. In practice, however, a PHA team may choose to analyze the Event Scenario having the same initiating cause and LOPC that is judged to have the most severe Consequences based on the particular circumstances and experience / knowledge of the team.

It is important to evaluate Consequences under the assumption that Critical Mitigation Elements (CMEs) and Key Mitigation Elements (KMEs) that may be in place to prevent initiation of the event or escalation of the event do not function as intended. This approach should then lead to the credible worst-case Consequences for the Event Scenario.

Consequences can be characterized by damage or harm in the form of injury to people, environmental impairment, property damage, loss of production revenue, or stress on community / agency relations that impacts the ability to operate. For each of the applicable categories of Consequence, PHA teams are required to estimate the Consequence level (severity) through the use of the OOG Risk Matrix.

The PHA team should also be cognizant of credible knock-on effects / escalation that can occur from the Consequence. For example, a jet fire from an ignited flange leak might cause an adjacent condensate line to rupture resulting in a larger fire that could have more severe Consequences such as property damage or environmental harm.

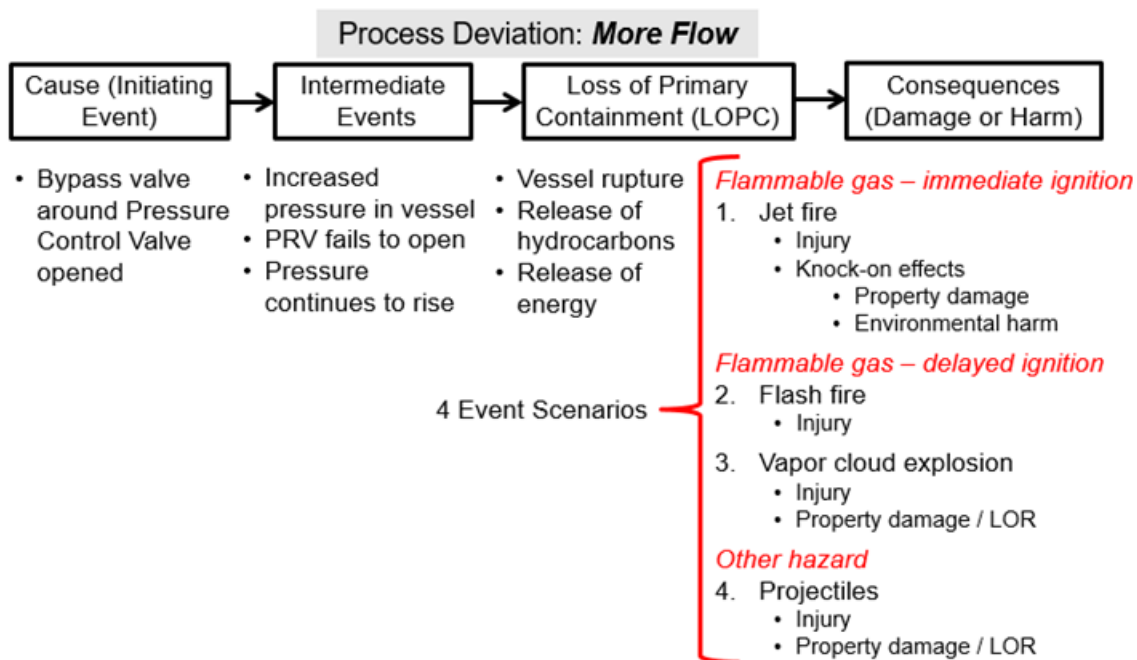


Figure 2: Example of LOPC Consequence Analysis

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4.3 Absolute Worst Case vs. Credible Worst-Case Consequences

If an Event Scenario progresses to the point of possible harmful consequences (without crediting safeguards), an absolute worst-case Consequence is considered a level of harm that, while still possible, is unlikely. In contrast, a credible worst-case Consequence is considered a level of harm that is likely should the Event Scenario progress to the point of causing harm. PHA teams should employ reasonable judgement when assigning the ultimate Consequence for an Event Scenario to generate meaningful results that represent a credible worst-case, rather than the absolute worst-case.

As an example, the gas dispersion model for an Event Scenario involving a hydrocarbon release of up to one hour finds that the released hydrocarbon gas is in the flammable range at a part of the Facility where only one person is stationed and conducts a routine operational activity. Though possible, it is not reasonable to assume that the release would occur when a group of four or more persons, who do not work in the area, happen to stop-by for an infrequent, non-routine permit sign-off. In this case, the Human Injury at a Facility Consequence per the OOG Risk Matrix should be Level 3 which equates to one Permanent Incapacitating Injury (PII), not Level 5 (more than three PIIs).

4.4 Propagation of Flammable Material Releases into Secondary Events

Process parameters such as pressure and flammable material properties, as well as the location of release, can affect the types of scenarios that could develop from a flammable material LOPC. Per common industry convention, PHA teams should generally assume that a flammable material release will eventually ignite due to presence of fired equipment, hot surfaces, or spark hazards such as from improper hazardous area classification and maintenance, thereof.

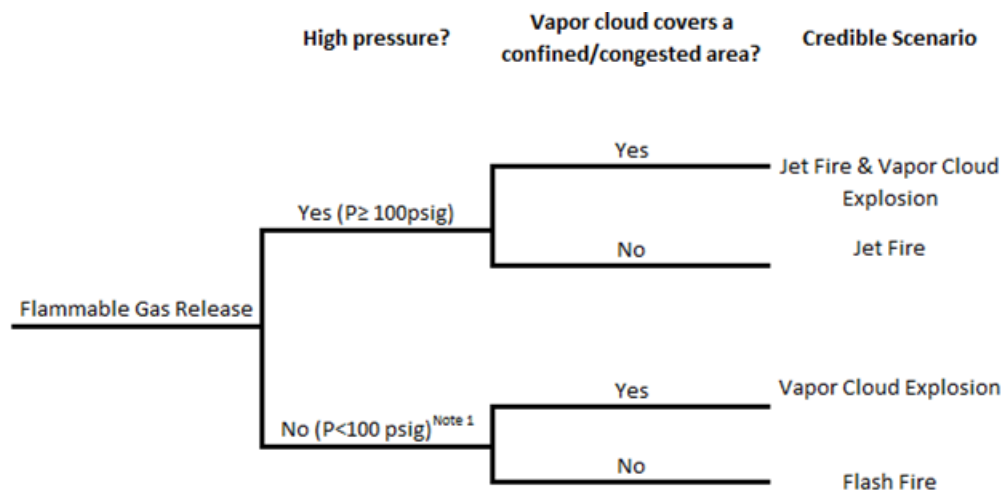
4.4.1 Flammable Gas LOPC

A flammable gas release can cause various hazardous effects such as a flash fire, a vapor cloud explosion, and/or a jet fire.

A vapor cloud explosion can occur from ignition of a flammable gas release within confined areas or in congested process areas where the flame front can accelerate through the confined/congested areas and develop damaging blast over-pressures. Generally, ignition of a gas release from a low-pressure system occurring in an open, unconfined and non-congested area such as a simple tank farm will probabilistically lead to only a flash fire.

A jet (torch) fire requires a relatively high system pressure to sustain the jet. As the process system pressure increases, a substantial jet fire from the leak area could be expected. For assessment purposes, jet fires are assumed to be orientated horizontally and can have both direct impingement and radiation effects on the receptor.

Figure 3: Flammable Gas Release Event Tree depicts the various credible events possible for a flammable gas release.

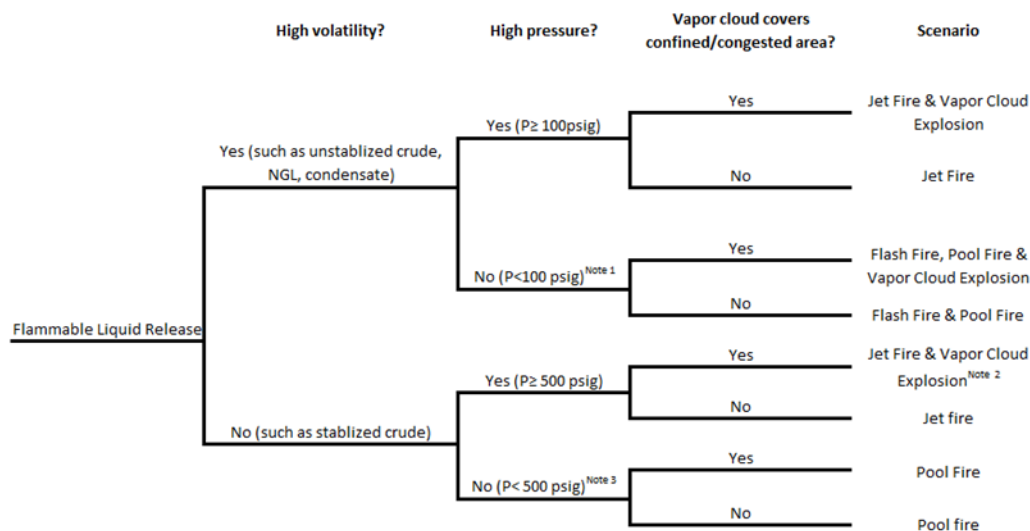


Note 1: Less than 100 psig discharge pressure is not postulated to generate a significant jet fire due to low release momentum (based on dispersion modeling runs using PHAST)

Figure 3: Flammable Gas Release Event Tree

4.4.2 *Flammable Liquid LOPC*

A flammable liquid release could develop into a liquid jet fire or lead to a pool fire. As a secondary event, a flammable vapor cloud could also develop based on liquid evaporation leading to similar effects as mentioned above for flammable gases. Figure 4: Flammable Liquid Release Event Tree depicts the various credible events possible for a flammable liquid release.



Note 1: Less than 100 psig discharge pressure is not postulated to generate a significant jet fire due to low release momentum (based on dispersion modeling runs using PHAST)

Note 2: Due to high degree of aerosolization possible from the release of a low-volatility flammable liquid, a vapor cloud explosion is deemed sustainable

Note 3: A low-volatility liquid release at less than 500 psig discharge pressure is not postulated to form aerosol, thus substantially reducing vapor cloud potential

Figure 4: Flammable Liquid Release Event Tree

4.5 Establishment of Consequence for Near-Field Personnel vs. Fixed Structures

When LOPC is identified during a PHA study such as from vessel/piping overpressure or leaks, the personnel performing routine operational and maintenance surveillance at manned facilities are assumed to suffer an immediate injury due to proximity to the process equipment. These receptors are thus termed “Near-Field Personnel”. In PHAs, the impact to the Near-Field Personnel is the primary focus.

The PHA teams are generally not tasked to evaluate Consequences to occupants in fixed occupied structures such as buildings. For such structures, a facility siting study per the requirements of OOG Procedure 60.400.310 and OOG Guideline 60.400.308.01, is the most appropriate tool to establish the credible Consequence to the occupants. However, if a facility siting study for occupied buildings has not been conducted, the PHA team should record the proposed impact scenario and capture a recommendation to do further follow-up such as a full siting study or more targeted dispersion modeling, as appropriate.

Similarly, if public Permanent Occupied Structures such as a residence or a business are in the vicinity of the system and the PHA team deems that a LOPC scenario could affect such offsite receptors, the team should record the scenario and capture a recommendation to do a follow-up study.

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To readily establish conclusive Consequence Levels within a PHA session, the OOG Radius of Exposure (ROE) Tool may be useful if the system release parameters are within the ROE Tool's specifications.

4.6 Manned Facility and Routine Presence

U.S. Occupational Safety & Health Administration (OSHA) considers a Facility unmanned if the personnel are at the site for an average of less than 1.5 man-hours per day and less than a total of 14.5 man-hours per week. These two separate thresholds must be used to establish if the Facility as a whole is manned. If a Facility is not manned, the Human Injury at a Facility category would not be considered credible as per guidance in OOG Guideline 60.400.301.03. An exception would be a scenario that involves action(s) performed by personnel within the affected area resulting in an acute Consequence to the personnel; such a case still warrants an analysis of the Human Injury at a Facility category at the unmanned Facility. An example of such a scenario is that during a routine site visit to an unmanned Facility, an operator erroneously leaves a bleed valve cracked open causing a vapor cloud to rapidly form leading to a fire. As this event has potential to cause immediate injury, Human Injury at a Facility category is applicable.

Establishment of Manning Levels & Definition of "Routine"

The Facility manning level is to be established on a cumulative basis for personnel that are routinely present. Routinely refers to normal operational surveillance plus any typical maintenance personnel supporting general activities such as preventive maintenance checks. For example, routinely is not meant to include maintenance overhauls, intervention activities or turnarounds. As merited based on anticipated Consequences/Risk level, these latter activities may require a separate risk assessment.

Agreement among the PHA team based on site operational history is deemed sufficient to establish if manning thresholds are met. It is generally not necessary to conduct manning surveys.

4.7 Human Injury at a Facility and Related Considerations

The Human Injury at the Facility category is dependent upon several factors such as personnel location and frequency of personnel's interaction with the process for routine surveillance. For example, in a central part of a manned Facility (such as a gas plant), based on release size, a Consequence Level 3 (one PII) may be assigned for Near-Field Personnel if a serious injury is possible based on LOPC. This is because central parts of a Facility are generally high traffic areas, with high likelihood of personnel presence in close proximity to the process.

In non-central areas of a manned Facility that are remote and have lower personnel presence (such as a tank farm), the Human Injury at the Facility category must still be analyzed. However, the IF-8 CME (Contingent Element) may be applied for Probability reduction if the personnel are present less than 10% of time.

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Elevating Consequence Levels to 4 and 5

Certain Facilities may routinely have more than one person present at a location that is in close proximity to the process where multiple PIIs could occur. In this case, assignment of a higher Consequence level may be appropriate. To consider more than one PII credible for an Event Scenario, thereby escalating the Consequence Level to 4 or 5, the PHA team should confirm that each additional person routinely spends a minimum of 1.5 hours/day or 14.5 hours/week on average in addition to the routine operational/maintenance surveillance hours spent by the single baseline one PII (Consequence Level 3) assumed for the first personnel.

For example, in a particular Gas Plant, one operator routinely surveils the process area based on workload and available level of automation, with limited routine maintenance personnel work. At this Facility, Consequence Level 3 (one PII) would be assigned to an Event Scenario of LOPC of toxic gas due to a seal failure on a compressor based on a deadhead condition. At another Facility, two operators routinely surveil the process area. Therefore, before a Consequence Level of 4 for the same Event Scenario is applied, it should be substantiated that both operators are routinely present for more than 1.5 man-hours per day or 14.5 man-hours per week.

4.8 Minimum Recommended Consequence Levels for Near-Field Personnel

In the context of this guideline, Near-Field Personnel are considered people who are proximate to the postulated source of harm and are likely to be impacted by an Event Scenario as they would not have time or ability to escape the impacts of events resulting from release of materials (LOPC), mechanical energy, or other sources of harm. Distant Personnel are considered people who are generally farther from a Hazard such that they would likely have time to react to an event and are less likely to be impacted by the Event Scenario.

In various sections, this guideline provides recommendations on equivalent release hole sizes for common failure modes of process vessels, tanks, piping and pipelines. Together with hole size, the material properties (flammability, toxicity), pressure, temperature and release elevation can be used to model the extent of impact. Generally, it should be assumed that the release occurs at the same elevation as Near-Field Personnel, directly impacting them, unless a hazardous release is specifically assumed to emanate from an elevated source such as a flare stack. Another important consideration for flammable material release consequences is the degree of nearby congestion/confinement which could lead to an explosion.

Although the consequence levels recommended in this section should apply in most cases, deviation from these levels may be justified based upon site-specific conditions and when supported by technical analysis. The types of release that could potentially affect Near-Field Personnel can be categorized into:

4.8.1 Flammable Material Release in Congested/Confined Areas

Subsequent to a flammable material release, a flash fire, jet fire, or explosion could reasonably affect personnel routinely present in a central area of a process Facility due to high probability of personnel present at close proximity in confined/congested areas.



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A Consequence Level 3 (one PII) for Human Injury at a Facility category should be recorded for Near-Field Personnel. If it is reasonable that more personnel could be affected based on manning level, a Consequence Level 4 or 5 could be applicable (see Section 4.7 above).

4.8.2 Flammable Material Release in Open Areas

For small leaks (hole size $\leq 1/4$ -inch diameter) occurring in relatively open areas of the Facility such as a simple tank farm, the released material would likely disperse without ignition. However, a flash fire could still develop in some situations, depending upon the fluid properties and atmospheric conditions. Accordingly, a lower Consequence Level 2 or 1 for Human Injury at a Facility category could be more appropriate for the Near-Field Personnel in such situations. A larger postulated release size may warrant a higher Consequence Level.

4.8.3 Toxic Gas Releases

Depending on toxicity level and concentration, a toxic gas cloud could travel a large distance. For toxic gas released at or above the level that is Immediately Dangerous to Life or Health (IDLH)¹, a Consequence Level 3 (one PII) for Human Injury at a Facility category should be recorded for Near-Field Personnel. If more personnel could be affected based on manning level, a Consequence Level 4 or 5 could be applicable (see Section 4.7 above). The impact for Distant Personnel should be supported by a dispersion modeling study.

4.8.4 Projectile Impact

Projectiles resulting from explosions or release of energy can present an impact hazard to people and property. Due to variability and difficulty in accurately analyzing possible projectile impacts as a result of explosions, these effects are not normally considered in a PHA. For Event Scenarios involving projectiles launched from release of energy such as failures involving reciprocating compressors and turbo-machinery, the near-field impact distance should be qualitatively estimated giving consideration to the likely projectile direction / orientation, possible shielding from other equipment and structures, potential or kinetic energy, and previous incident / comparative data as available. However, projectile distance calculations and probabilistic impact studies are considered beyond the scope of Consequence estimation for PHA Event Scenarios.

A Consequence Level 3 (one PII) for Human Injury at a Facility category should be recorded if there are areas considered manned within the near-field impact distance. If more personnel could be affected based on manning level, a Consequence 4 or 5 could be applicable (see Section 4.7 above).

¹ In the specific case of hydrogen sulfide (H₂S), an exposure level of 700 ppm is associated with a PII for Human Injury at a Facility. Refer to OOG 60.400.301.01, Guideline for Assigning H₂S and CO₂ Consequence Levels.

4.9 Pressure Vessel Overpressure Considerations

This section provides guidance on hole sizes to consider for overpressure event scenarios based on the level of overpressure to which a pressure vessel could be subjected.

The ASME Boiler and Pressure Vessel Code Section VIII uses the concept of Maximum Allowable Working Pressure (MAWP) which provides a safe upper pressure limit for normal operation. This pressure is determined based on the design, temperature, allowable stress for each material in the system, and the installed thickness minus the corrosion allowance. ASME Code incorporates a significant vessel design margin to help ensure safe operation. For carbon and stainless steel materials common to Oxy's production equipment, this design margin is 3.5:1 when compared to the minimum material tensile strength and about 2:1 compared to the minimum material yield strength. Per the current ASME Code, a new vessel must be hydrotested at a pressure of 1.3 x MAWP to confirm sound vessel construction.

For pressure vessels protected by a single pressure relief device (PRD), the relief setting must be at or lower than the MAWP. The Code allows the pressure in a vessel (accumulated pressure) to exceed the MAWP during upset and fire relieving conditions to account for the characteristics of the PRD as it opens to full relieving capacity and to allow operational flexibility. Using the 2015 ASME code as an example, a single relief valve must limit the maximum vessel pressure to no more than 110% of MAWP. On vessels having multiple relief valves, the maximum pressure must be limited to no more than 116% of MAWP. In the case of vessel exposure to fire, the pressure relieving capacity of single or multiple relief valves must be large enough to limit the maximum pressure to no more than 121% of MAWP.

Accordingly, taking the design margins that the Code provides for safety into account and recognizing other factors that may degrade these margins, Table 1, below presents recommended equivalent release hole size diameters of LOPC events based on overpressure conditions for a pressure vessel.

Table 1: Recommended Release Hole Size for Metallic Pressure Vessel Overpressure

Pressure	Significance	Consequence	Recommended Hole Size
$\leq 1.1 \times \text{MAWP}$	Allowable accumulation for process upset cases (<u>non-fire</u>) protected by a <u>single</u> relief device	No expected Consequence at this accumulation level	Not applicable, no LOPC expected
$> 1.1 \times \text{MAWP}$ up to $1.3 \times \text{MAWP}$	Standard hydrostatic test pressure at $1.3 \times \text{MAWP}$	Flange leak or instrumentation leak	1/4-inch (6 mm)
$> 1.3 \times \text{MAWP}$ up to $2 \times \text{MAWP}$	Stresses in vessel above original proof test up to yield point.	Large flange leak or threaded connection failure	3/4-inch (20 mm)

Pressure	Significance	Consequence	Recommended Hole Size
> 2 x MAWP	Stresses greater than yield strength	Potential for permanent deformation and vessel rupture	6-inches (150 mm)

Note: Assumptions made in the above table are that the vessel is designed/tested per ASME Code, the current vessel condition meets the design intent, and the vessel is maintained through the facility mechanical integrity (MI) program. If there is no documentation to demonstrate the integrity of the vessel, then any scenario that imposes a pressure greater than 1.1 x MAWP should be assumed to result in a vessel rupture.

Non-Code pressure vessels are commonly found in field installations and generally have little or no documentation to support their claimed MAWP. They are typically built with a lower design factor but are generally hydrostatically pressure tested to 1.3 x MAWP, similar to Code vessels. When documentation is available to support the integrity of such vessels, a vessel rupture case would be appropriate at an overpressure greater than 1.3 x MAWP. Otherwise, a scenario that imposes a pressure greater than 1.1 x MAWP should be assumed to result in a vessel rupture.

4.10 Process Piping Overpressure Considerations

ASME B31.3 is the relevant Code employed for process piping design. This piping design Code uses the concept of design pressure, and in accordance, process piping is generally hydrotested at a pressure of 1.5 times the design pressure.

Table 2 presents recommended equivalent release hole size diameters based on overpressure failures for metallic gas and liquid process piping. The recommended hole size for rupture is limited to 6-inches for liquid service piping due to non-compressible fluid characteristics that rapidly dissipates pressure upon initial release, thus reducing the likelihood of a much larger release size. Note: Two-phase fluid service should be treated the same as for gas piping.

For non-metallic piping, due to lack of inspection capability, a full-bore rupture for an overpressure condition above the design pressure should be assumed.

In general, piping systems constructed to ASME B31.3 incorporate a design factor of 3:1 when compared to minimum material tensile strength and about 2:1 compared to the minimum material yield strength. For conservatism, the threshold for complete rupture has been set at 2 x Design Pressure (DP), similar to pressure vessels.

Table 2: Recommended Release Hole Size for Metallic Process Piping Overpressure

Pressure	Significance	Consequence	Recommended Hole Size
$\leq 1.1 \times$ Design Pressure	Allowable accumulation for process upset cases (<u>non-fire</u>) protected by a <u>single</u> relief device	No expected Consequence at this accumulation level	Not applicable, no LOPC expected
$> 1.1 \times$ Design Pressure up to $1.5 \times$ Design Pressure	Standard hydrostatic test pressure per ASME B31.3	Flange leak or instrumentation leak	1/4-inch (6 mm)
$> 1.5 \times$ Design Pressure up to $2 \times$ Design Pressure	Stresses in piping above original proof test up to yield point	Large flange leak, threaded connection failure or piping crack	3/4-inch (20 mm)
$> 2 \times$ Design Pressure	Stresses greater than yield strength	Potential for permanent deformation and piping rupture	Gas: Full-bore rupture Liquid: Full-bore rupture up to 6-inches (150 mm)

Note: Assumptions made in the above table are that the process piping is designed/tested per relevant codes, the current condition meets the design intent, and the piping is maintained through the facility MI program. A full-bore rupture failure should be assumed when pressure exceeds $1.1 \times$ design pressure if there is no supporting data to demonstrate the integrity of the piping.

4.11 Pipeline Overpressure Considerations

B31.4 (liquid pipelines) and B31.8 (gas pipelines) are the most common codes employed for pipeline design. In general, pipelines constructed to ASME B31.4 and B31.8 incorporate a design factor of about 2:1 when compared to minimum material tensile strength and about 1.4 - 1.6:1 compared to the minimum material yield strength. ASME B31.8 establishes more conservative design factors for gas pipelines in locations with a higher number of occupied buildings. While the greater design factors for these locations² (Location Classes 3 and 4) could justify a higher pressure criteria for leaks and pipeline rupture, for conservatism and practical application the recommended hole sizes are generally based on the lower design factors associated with Location Classes 1 and 2.

Table 3 presents recommended equivalent hole size diameters based on overpressure failures for both metallic gas and liquid pipelines. The recommended hole size for rupture is limited to 6-inches for liquid service piping due to non-compressible fluid characteristics that rapidly

² Refer to ASME B31.8 Gas Transmission and Distribution Piping Systems, section 840.2.2 for a description of Location Classes.

dissipates pressure upon initial release, thus reducing the likelihood of a much larger release size.

For non-metallic pipelines, due to lack of inspection capability and the typical failure mode, a full-bore rupture for an overpressure condition above the design pressure should be assumed. Typical non-metallic materials for produced water, oil, and gas service pipelines are High Density Polyethylene (HDPE) constructed to ASTM 4710, composite materials (e.g., HDPE and wound fiberglass), fiberglass-reinforced plastic (FRP) and glass-reinforced plastic (GRP).

Above-ground and below-ground pipelines should be treated the same in terms of the overpressure scenarios listed. Also, a two-phase pipeline should be treated the same as a gas pipeline for the purpose of this section.

Table 3: Recommended Release Hole Sizes for Metallic Pipelines Overpressure

Pressure	Significance	Consequence	Recommended Hole Size
≤ Design Pressure	Pressure range up to the MAOP	No adverse Consequences	Not applicable, no LOPC expected
>Design Pressure up to 1.25 x Design Pressure	Standard hydrostatic test pressure per ASME B31.4 and B31.8 (Location Classes 1 & 2)	Flange leak or instrumentation leak	1/4-inch (6 mm)
> 1.25 x Design Pressure up to 1.5 x Design Pressure	Stresses above original hydrostatic test up to the approximate yield point	Large flange leak, threaded connection failure or piping crack	3/4-inch (20 mm)
> 1.5 x Design Pressure	Stresses greater than yield strength	Potential for deformation and piping rupture	Gas: Full bore rupture Liquid: Full bore rupture up to 6-inches (150 mm)

Note: Assumptions made in the above table are that the pipeline is designed/tested per relevant codes, the current condition meets the design intent, and the vessel is maintained through the facility mechanical integrity (MI) program. A full-bore (up to 6-inches) rupture failure should be assumed when pressure exceeds 1.1 x design pressure if there is no supporting data to demonstrate the integrity of the pipeline.

4.12 Corrosion, Erosion, & Other Damage Mechanisms

The following damage mechanisms are often covered in a PHA or task risk assessment:

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- Internal / external corrosion or erosion
- Vehicular, dropped load or other impact to above ground components
- Excavation strike leading to potential pipeline failure

The intent of reviewing these damage mechanisms is to identify areas of processes and equipment that may be uniquely susceptible to harm and that may not be sufficiently managed by the relevant programs such as mechanical integrity, maintenance, critical lifts, and excavation. For regulated facilities (gas plants, CO₂ recovery plants, offshore platforms), these programs are often mature and functioning well. In such cases, the PHA team should assess whether the damage mechanism presents a specific hazard that would not be controlled through these programs based on their knowledge and prior experiences / incidents involving the system under study and for similar systems.

If the damage mechanisms are believed to be adequately managed by these programs, then the PHA worksheets should indicate that no specific / new hazards or issues were identified. If a unique situation or hazard exists that is not sufficiently managed by these programs, then the PHA team should develop and analyze potential LOPC Event Scenarios based on the guidance provided in this section. Likewise, facilities that do not have strong programs to manage these damage mechanisms or that have identified specific / new hazards should develop and analyze potential LOPC Event Scenarios utilizing this guidance.

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Table 4 presents recommended release hole sizes for pipelines for the common damage mechanism based on past industrial accidents and Oxy's historical experience. Scenarios 1 – 4 cover corrosion / erosion damage mechanisms applicable to typical carbon steel materials, though erosion scenarios may also apply to non-metallic materials. Use of higher alloy and stainless steels may warrant use of smaller release hole sizes than recommended in this table.

Recommended release sizes are generally larger for equipment containing gases compared with liquids due to the higher stored energy within compressed gases that does not immediately dissipate upon a breach in the pressure-containing walls. Upon release, the stored energy tends to open-up the breach. Additionally, smaller release sizes are recommended for lower-pressure corrosion / erosion scenarios (typically Class 150 piping) because the lower contained energy does not support large failures in either gas or liquid services.





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Table 4: Recommended Release Hole Size for Pipeline Scenarios

Pipeline Release Scenario	Conditions / Technical Basis	Potential Consequences	Recommended Hole Size
Scenario 1. Internal or External Corrosion / Erosion; with MI program ³	MI monitoring of buried pipelines is in place, but possible undetected corrosion or damage not identified between inspections	Localized damage; small or very small release size	Gas or Liquid: 1/2-inch (12 mm)
Scenario 2. Internal or External Corrosion / Erosion, without MI program; ASME Class 300 and higher	MI monitoring is not in place; no ILI / GUL / partial excavation (direct assessment) of buried lines Breach in pressure-containing wall tends to "open up" under high gas pressure but less so in liquid service	General wall loss; large release size for gas service and small release size for liquid service	Gas: Full bore release up to 6-inches (150 mm) Liquid: Full bore release up to 2-inches (50 mm)
Scenario 3. Internal or External Corrosion / Erosion, without MI program; ASME Class 150 or lower	MI monitoring is not in place; no ILI / GUL / partial excavation (direct assessment) of buried lines Lower pressure service lacks the energy needed to cause large failures	General wall loss; small release size	Gas or Liquid: Full bore release up to 2-inches (50 mm)
Scenario 4. Internal or External Corrosion / Erosion, with internal liner or coating	Monitoring of internal pipe liner or coating is generally not possible; potential for liner damage during fabrication / construction resulting in localized corrosion / erosion	Localized corrosion / erosion damage; small release size	Gas or Liquid: 1/2-inch (12 mm)
Scenario 5. Vehicular Impact, High Momentum	Impact by light/medium duty pickup truck: gross vehicle weight up to 10,000 lbs. (4,500 kg) traveling at 55 mph (90 kph) or faster; large truck at any speed above 10 mph (15 kph)	Puncture or rupture of pressure-containing wall; up to large release size	Gas or Liquid: Largest piping size up to 6---inches (150 mm)

³ Scenario 1 should be considered only when an MI program is in place for buried pipelines (ILI / GUL / partial excavation - direct assessment) but the effectiveness is not well established, or if there is a potential downgraded condition or known anomaly.





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Pipeline Release Scenario	Conditions / Technical Basis	Potential Consequences	Recommended Hole Size
Scenario 6. Vehicular Impact, Medium Momentum	Impact by light/medium duty pickup truck: gross vehicle weight up to 10,000 lbs. (4,500 kg) traveling at 20 – 50 mph (30 – 80 kph)	Puncture or rupture of pressure-containing wall; up to medium release size	Gas or Liquid: Largest piping size up to 2-inches (50 mm)
Scenario 7. Vehicular Impact, Low Momentum	Impact by light/medium duty pickup truck ⁴ : gross vehicle weight up to 10,000 lbs. (4,500 kg) traveling at 15 mph (25 kph) or slower	Puncture or rupture of pressure-containing wall; up to small release size	Gas or Liquid: 3/4-inch (20 mm)
Scenario 8. Heavy Dropped Load	Heavy dropped load is considered more than 20,000 lbs. (9,000 kg) from a height of 20 feet (6 m)	Puncture or rupture of pipeline; up to large release size	Gas or Liquid: Largest piping size up to 6-inches (150 mm)
Scenario 9. Medium Dropped Load	Medium dropped load is considered 5,000 to 20,000 lbs. (2,300 to 9,000 kg) from a height of 20 feet (6 m)	Puncture or rupture of pipeline; up to medium release size	Gas or Liquid: 2-inch (50 mm)
Scenario 10. Light Dropped Load	Light dropped load is considered less than 5,000 lbs. (2,300 kg) from a height of 20 feet (6 m)	Puncture or rupture of pipeline; up to small release size	Gas or Liquid: 3/4-inch (19 mm)
Scenario 11. Excavation Line Strike, Underground metallic pipeline	Typical powered excavating equipment: earth auger, backhoe, motor grader	Puncture or rupture of pipeline; up to medium release size	Gas or Liquid: 2-inches (50 mm)
Scenario 12. Excavation Line Strike, Underground non- metallic pipeline	Typical powered excavating equipment: earth auger, backhoe, motor grader Larger recommended release size than for metallic pipelines due to lower expected puncture / tear resistance	Puncture or rupture of pipeline; up to large release size	Gas or Liquid: - Largest piping size up to 6-inches (150 mm)

⁴ Scenarios involving larger weight vehicles travelling at low speeds should be considered medium or high momentum impacts.





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Table 5 presents recommended release hole sizes for corrosion / erosion and impact scenarios that may be encountered in PHAs that pertain to piping and pressure vessels. The recommended hole sizes are the same as for pipelines but are shown separately in this table due to differences in the conditions / technical basis.

Table 5: Recommended Release Hole Size for Piping and Pressure Vessel Scenarios

Release Scenario	Conditions / Technical Basis	Potential Consequences	Recommended Hole Size
Scenario 1. Internal or External Corrosion / Erosion; with MI program ⁵	MI monitoring is in place but possible undetected corrosion (e.g., corroded small bore pipe threads) or damage not detected between inspections	Localized damage; small or very small release size	Gas or Liquid: 1/2-inch (12 mm)
Scenario 2. Internal or External Corrosion / Erosion, without MI program; ASME Class 300 and higher	MI monitoring is not in place. Breach in pressure-containing wall tends to “open up” under high gas pressure but less so in liquid service	General wall loss; large release size for gas service and medium release size for liquid service	Gas: Full bore release up to 6-inches (150 mm) Liquid: Full bore release up to 2-inches (50 mm)
Scenario 3. Internal or External Corrosion / Erosion, without MI program; ASME Class 150 or lower	MI monitoring is not in place. Lower pressure service lacks the energy needed to cause large failures	General wall loss; medium release size	Gas or Liquid: Full bore release up to 2-inch
Scenario 4. Internal or External Corrosion / Erosion, with internal liner or coating	Monitoring of internal pipe liner or coating is generally not possible; potential for liner damage during fabrication / construction resulting in localized corrosion / erosion	Localized corrosion / erosion damage; small release size	Gas or Liquid: 1/2-inch (19 mm)
Scenario 5. Vehicular Impact, High Momentum	Impact by light/medium duty pickup truck: gross vehicle weight up to 10,000 lbs. (4,500 kg) traveling at 55 mph (90 kph) or faster; large truck at	Puncture or rupture of pressure-containing wall; up to large release size	Gas or Liquid: Largest piping size up to 6-inches (150 mm)

⁵ Scenario 1 should be considered only when an MI program is in place for piping and vessels, but the effectiveness is not well established, or if there is a potential downgraded condition or known anomaly.





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Release Scenario	Conditions / Technical Basis	Potential Consequences	Recommended Hole Size
	any speed above 10 mph (15 kph)		
Scenario 6. Vehicular Impact, Medium Momentum	Impact by light/medium duty pickup truck: gross vehicle weight up to 10,000 lbs. (4,500 kg) traveling at 20 – 50 mph (30 – 80 kph)	Puncture or rupture of pressure-containing wall; up to medium release size	Gas or Liquid: Largest piping size up to 2-inches (50 mm)
Scenario 7. Vehicular Impact, Low Momentum	Impact by light/medium duty pickup truck ⁶ : gross vehicle weight up to 10,000 lbs. (4,500 kg) traveling at 15 mph (25 kph) or slower	Puncture or rupture of pressure-containing wall; up to small release size	Gas or Liquid: 3/4-inch (19 mm)
Scenario 8. Heavy Dropped Load	Heavy dropped load is considered more than 20,000 lbs. (9,000 kg) from a height of 20 feet (6 m)	Puncture or rupture of pressure-containing wall; up to large release size	Gas or Liquid: Largest piping size up to 6-inches (150 mm)
Scenario 9. Medium Dropped Load	Medium dropped load is considered 5,000 to 20,000 lbs. (2,300 to 9,000 kg) from a height of 20 feet (6 m)	Puncture or rupture of pressure-containing wall; up to medium release size	Gas or Liquid: 2-inch (50 mm)
Scenario 10. Light Dropped Load	Light dropped load is considered less than 5,000 lbs. (2,300 kg) from a height of 20 feet (6 m)	Puncture or rupture of pressure-containing wall; up to small release size	Gas or Liquid: 3/4-inch (19 mm)

Table 6 presents recommended release hole sizes for corrosion / erosion and impact scenarios commonly covered in PHAs for metallic and non-metallic tanks.

Table 6: Recommended Release Hole Sizes for Atmospheric Tanks

Release Scenario	Conditions / Technical Basis	Potential Consequence	Recommended Hole Size
Scenario 1. Internal or external corrosion or erosion in carbon	Localized damage not observed / corrected between inspections	Very small release size	1/4-inch (6 mm)

⁶ Scenarios involving larger weight vehicles travelling at low speeds should be considered medium or high momentum impacts.



Release Scenario	Conditions / Technical Basis	Potential Consequence	Recommended Hole Size
steel tank with MI program ⁷			
Scenario 2. Internal or external corrosion or erosion in carbon steel tank without an MI program	General thinning of floor or wall plates	General wall loss leading to small release size	3/4-inch (20 mm)
Scenario 3. Coating or lining failure in tank	Holiday in coating or lining creates localized damage to tank floor or walls	Very small release	1/4-inch (6 mm)
Scenario 4. Impact to carbon steel tank	Based on typical low momentum vehicle impact	Deformation of tank wall; potential for puncture; medium release size	2-inches (150 mm)
Scenario 5. Impact to non-metallic tank	Based on typical low momentum vehicle impact to tank	Tearing / breaking of tank wall; large release size	6-inches (150 mm)

4.13 Gas Blow-by Damage to Low-Pressure Tanks

Generally, low-pressure fixed roof tanks are employed in the oil & gas industry for handling and storage of oil and produced water. In terms of design of these tanks, API 12D, API 12F, API 650, API 620, BS EN 12285, and BS EN 14015 are the most prevalent codes. These tanks are generally susceptible to “gas blow-by” pressure from an upstream system operating at a significantly higher pressure initiated by a potential loss of level in the upstream system boundary vessel.

The gas blow-by rate from the upstream vessel should be estimated based on the pressure differential among the highest postulated operating pressure of the upstream system and internal pressure rating of the downstream tank accounting for discharging coefficients (CVs) of any control valves in between, as well as piping friction losses.

4.13.1 Hole Size and Release Rate Estimation for a Tank Without a Frangible Roof

For a gas blow-by scenario involving low-pressure fixed roof tank without a frangible roof, a release hole size equivalent of 6 inches diameter is recommended because significant damage to a tank is generally possible. This hole size estimate should be used to establish the gas discharge rate from the tank to the atmosphere at the rated pressure

⁷ Scenario 1 should be considered only when an MI program is in place for tanks, but the effectiveness is not well established, or if there is a potential downgraded condition or known anomaly.

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of the tank. As an alternative more conservative approach, the calculated gas blow-by rate, could be conservatively applied directly as the gas release rate to the atmosphere.

The gas release should be conservatively assumed to occur at ground level in a horizontal direction to develop any dispersion modeling-based radius of exposure, as required.

4.13.2 Hole Size and Release Rate Estimation for a Tank with a Frangible Roof

It is not generally practicable to design low-pressure tank vents to relieve internal fire, explosion or chemical reaction or “roll over” effects caused by mixing of layers having different temperatures or compositions. These cases have exceptionally high relief demand flowrates. In these cases, it may be necessary to specify a tank design having a frangible roof or weak roof to shell seam in order to minimize the amount of damage to the tank. The frangible roof is designed to “open up” at a lower pressure than the tank shell or floor. Storage tanks compliant with API standards have frangible roof connection as a standard feature.

Under a gas blow-by case, it should be expected that a tank with a frangible roof will open up at the top to allow the incoming gas to discharge to atmosphere through a proportionately sized hole at the gas blow-by rate. In this case, the gas release to the atmosphere should also be conservatively assumed to occur in a horizontal direction at the tank roof level to develop any dispersion modeling-based radius of exposure, as required. Consult API Publication 937 for the design of a frangible roof-to-shell attachment.

4.13.3 Verification of Potential Impacts from a Mitigated Tank Release

Low-pressure tanks generally have vents and hatches that are designed to reduce excess pressure before tank damage can occur. In all cases, dispersion modeling should be performed to ensure that the discharge of a hazardous gas through the vent/hatch in an emergency relief case does not pose any hazardous ground level concentrations or impact to any nearby receptor.

4.14 Vacuum Damage to Pressure Vessels and Tanks

A vacuum condition is generally created by rapid unloading out of the vessel/tank or a cool down of the contents due to ambient conditions. The mechanism of vacuum failure is different from internal over-pressure failure. Internal pressure failure in gas service typically results in a violent rupture or large tearing open of the vessel/tank. In contrast, during vacuum failure (external pressure) the vessel/tank can no longer support its shape and suddenly, irreversibly collapses to a new lower volume shape.

Supported by industrial accident data, vacuum damage to pressure vessels can cause significant deformation, however, with no or minimal loss of containment of contents. As a result, it is reasonable for a PHA team to assume that a vacuum related LOPC scenario will lead to lower effective hole release size as compared to vessel/tank overpressure failures. Accordingly, a hole size of 1/4” for pressure vessels and 3/4” for tanks is recommended for vacuum collapse



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scenarios, contingent upon the source of vacuum being removed following collapse to allow entrapped liquids to drain or for possible subsequent equipment pressurization that causes release of material.

A special case of vacuum design is a pressure vessel rated for full vacuum (and maintained as such). For such vessels, the LOPC due to a vacuum condition should be deemed non-credible.

4.15 Overpressure Damage from or To Pumps and Compressors

There are two main classes of rotating equipment that exhibit different modes of excess pressure generation: positive displacement and centrifugal machinery.

4.15.1 Overpressure via Positive Displacement Equipment

The most common types of positive displacement machinery used in process industry are reciprocating compressors and pumps, screw compressors, and rotary vane pumps and compressors. Two common scenarios analyzed in PHAs involving positive displacement equipment are:

1. Blocked outlet
2. Liquid carry-over (positive displacement compressor, only)

For these scenarios involving positive displacement compressors, it is typical for internal components to fail such as the crankshaft, connecting rod, piston rod, or valves without incurring an LOPC. It is also possible for external components like the driver coupling to fail or for the engine or motor driver to shut down or stall on excessive torque or amperage. Design factors that could potentially prevent an LOPC are described in the following section. There is still a reasonable probability that the higher pressure generated from positive displacement equipment in the blocked outlet scenario would result in an LOPC from that equipment or in downstream equipment. It should be assumed that the excessive pressure could cause a large flange leak or threaded connection failure for which a ½" hole size release would be appropriate⁸.

For a blocked outlet scenario of a positive displacement pump, the pressure can build to an excessively high level very quickly and cause pressure retaining components to fail or piping to rupture⁹. Therefore, a release hole size equal to the nominal discharge piping size should be assumed, up to a maximum of 6-inches hole size. Although positive displacement pumps are typically provided with an internal or external pressure relief device to protect against a blocked outlet scenario, the pressure relief device should be considered a CME and not a device that renders this scenario "non-credible".

Design Factors Potentially Preventing LOPC

⁸ OOG 60.400.301.10 Credible Consequences & CME Credits for Liquid Carryover Scenario to a Compressor – Position Statement recommends a ½" hole size as the maximum credible release size.

⁹ A blocked outlet of a pump or liquid carryover to a compressor can also result in a near-field projectile hazard which should be assessed per Section 4.8.4.

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There are two design factors applicable to positive displacement machinery that potentially could prevent an LOPC are:

1. Inability of the motor driver to deliver enough mechanical power to operate the pump/compressor at a sufficiently high outlet pressure to cause a pressure retaining component to fail; and,
2. Mechanical failure of the power transmission coupling (weaker component) due to excessive torque required by the compressor to produce a higher discharge pressure.

If the PHA team believes that either of the above design factors would render an LOPC event non-credible, then a qualified rotating equipment specialist should be consulted to evaluate the specific machinery and determine if the team's position is valid.

Note: There are generally safeguards incorporated into the design such as motor overcurrent protection, torque limiting devices, and interlocks to de-energize the driver before a mechanical or electrical failure. As per general Consequence development convention, these safeguards should not be credited when estimating the worst credible Consequences. However, if sufficiently robust, then they could be credited for reducing the probability of incurring an LOPC event (CME PR-20 Physical Control or Protection Devices).

4.15.2 Overpressure via Centrifugal Equipment

The two primary modes of excess pressure generation applicable are blocked outlet (deadheading) of pumps and compressors, and surge in compressors.

Blocked Outlet on Pumps and Compressors

Excessive pressure may be generated in piping and equipment downstream of centrifugal equipment if the flow is blocked such as by closing a downstream valve. This creates a condition commonly known as "deadheading".

Centrifugal equipment deadhead pressure can be determined via the use of a pump or compressor curve and can be compared with the downstream equipment pressure rating to determine if LOPC is plausible. The assumed holes size of a release should be based upon the percent overpressure and applicable guidance table in Sections 4.9 – 4.11.

Another form of centrifugal pump deadheading can occur in which both the suction and discharge valves are closed while the pump continues to operate with fluid within as a result of a startup or shutdown error. This condition can cause the temperature of the trapped fluid to increase to a point that the pressure from thermal expansion or vaporization of the fluid becomes extreme, potentially rupturing the pump. This scenario can result in both an LOPC hazard and a projectile hazard to persons in the area from components that could be thrown from the failed equipment (see Section 4.8.4).

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The LOPC size should be considered equivalent to the pump discharge pipe size up to 6" hole size. As this type of event could develop over a period of hours, a human injury Consequence should give consideration to the likelihood of persons being in the estimated near-field impact distance at the time of the event.

Surge in Centrifugal Compressors

Surge is an unstable operating mode that can occur at low flow rates such as may occur when the suction valve to a compressor is closed below a lower limit. It is characterized by large flow and pressure fluctuations in the machine that can result in severe machine damage to internal components. When such a condition is identified by a PHA team, failure of the compressor shaft seals should be considered, resulting in material released between the shaft (rotor) and seal gland. For typical compressors, a 1/2" to 3/4" hole size is recommended to represent an LOPC event from surge.

4.16 Pump Seal Failures & Leaks

Quite often PHA Teams identify scenarios related to pump seal failures involving hazardous material LOPCs for which appropriate Consequence Level must be identified. Scenarios that are commonly analyzed include:

- General mechanical seal failure (seal face wear, elastomer degradation, corrosion / fatigue failure of metal components)
- Abnormal operating conditions (leading to high vibration or temperature and possible seal failure)
 - Pump deadheaded (no flow condition)
 - Loss of barrier fluid from various causes (high temperature condition)
 - Cavitation (low suction pressure condition)
 - Flow recirculation¹⁰ / discharge cavitation (low flow condition)

There are various seal types and arrangements as described in API Standard 682, Mechanical Shaft Seals. For simplification, all seal types should be assumed to fail in a manner leading to an LOPC. However, dual-seal arrangements utilizing a barrier fluid to isolate the process fluid from the seal faces could be considered a safeguard (CME) to minimize the probability of an LOPC. When estimating spill release rates based on hole sizes provided in the following sections, the release pressure should be assumed to be the suction pressure of the pump unless the seal chamber is purged by a line from the discharge of the pump, in which case the discharge pressure would be appropriate. The seal chamber also may be pressurized by a barrier fluid (dual seal arrangement) to isolate the process fluid from the seal faces. In this case, the release pressure should still be considered the pump suction pressure, but credit could be given to the

¹⁰ When a centrifugal pump is operated at a low flow rate, a condition known as flow recirculation can occur in which there is a flow reversal at the inlet or at the discharge tips of the impeller vanes. This condition can cause cavitation and surging at the inlet vanes, resulting in excessive vibration that leads to seal damage, impeller damage, or shaft breakage over time.



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seal barrier fluid acting as CME. Credit would not be given to the seal barrier fluid if loss of that system was considered the initiating cause for seal failure.

Historically, pump seal failures initially develop as small leaks that are readily evident to personnel performing routine operational and maintenance surveillance. If such a failure is not identified and remediated at the early stage, leaks can intensify and become much larger. Accordingly, for facilities that are manned (thus more frequently monitored during operator rounds or maintenance activity), pump seal leaks would likely be stopped in a short period of time from when they start and when they are relatively small. Conversely, for pumping systems located at unmanned or remote locations of a Facility where the likelihood of early detection of pump seal failure is lower, a larger hole size assumption is appropriate.

4.16.1 Pump Seal Failure at a Manned Facility

Recommended hole sizes to represent relatively large pump seal failures at manned facilities are presented in Table 7. These values are based on ranges of pump shaft sizes (per API Standard 610, Table 6 seal chamber sizes) and a set assumed gap between stationary and rotating elements (diametrical clearance) of 0.5 mm (0.02 in.).

Table 7: Recommended Hole Size for Pump Seal Failure – Manned Facility

Shaft Diameter, Maximum in mm (inches in parentheses)		Hole Size Diameter in mm (inches in parentheses)
20 (0.79)	→	6 (0.25)
30 (1.18)	→	8 (0.30)
40 (1.57)	→	9 (0.35)
50 (1.97)	→	10 (0.39)
60 (2.36)	→	11 (0.43)
70 (2.76)	→	12 (0.47)
80 (3.15)	→	13 (0.50)
90 (3.54)	→	13 (0.53)
100 (3.94)	→	14 (0.56)
110 (4.33)	→	15 (0.58)

4.16.2 Pump Seal Failure at Unmanned or Remote Areas of a Facility

Recommended hole sizes to represent complete failure of pump seals at unmanned or remote areas of facilities are presented in Table 8. These values are based on ranges of pump shaft sizes (per API Standard 610, Table 6 seal chamber sizes) and an assumed gap between rotating elements (shaft or seal sleeve O.D.) and stationary seal components (typically seal gland or fixed bushing) that is equal to twice the minimum diametrical clearance per API Standard 682, Table 1. Consideration is not given to reduced clearance



that floating carbon bushings or throttle bushings provide as these parts may be damaged from severe pump vibration or wear.

Table 8: Recommended Hole Size for Pump Seal Failure – Unmanned or Remote Facility

Shaft Diameter, Maximum in mm (inches in parentheses)		Hole Size Diameter in mm (inches in parentheses)
20 (0.79)	→	13 (0.5)
30 (1.18)	→	15 (0.6)
40 (1.57)	→	18 (0.7)
50 (1.97)	→	20 (0.8)
60 (2.36)	→	22 (0.9)
70 (2.76)	→	33 (1.3)
80 (3.15)	→	36 (1.4)
90 (3.54)	→	38 (1.5)
100 (3.94)	→	40 (1.6)
110 (4.33)	→	42 (1.7)

4.17 Flare Hazard Scenarios

As a pre-requisite to a flare system PHA, it is recommended that a flare study be completed to ensure Consequence levels related to typical hazard scenarios can be appropriately analyzed. A typical flare study includes the ground level thermal radiation at maximum design-flow rates and the ground level concentration of flare gas in terms of Lower Flammability Limit (LFL) and toxic components (e.g. H₂S or CO₂), in the case of flame-out.

Common flare hazards include –

- Thermal radiation around flares with unrestricted access
- Liquid carryover to flare resulting in rain-out (burning liquid)
- Flame-out of flare resulting in ground-level flammable or toxic gas concentration
- Flare noise

4.17.1 Thermal Radiation – Flares with Unrestricted Access

For flares that do not have restricted access¹¹, the possibility of a burn injury may exist if a person is present in close proximity at ground level to a flare coincident with a large release of material to the flare such as activation of a pressure relief device or a process blow-down. In general, people could be expected to escape from the area but with the potential for burns due to exposure to thermal radiation. For this scenario, the

¹¹ Restricted access areas generally encompass locations where the potential thermal radiation level is 6.3 kW/m² or higher as specified in OOG Engineering Standard ES-10.400.502, Facility Layout and Spacing. Typically, these areas are fenced and have posted signs indicating that access is restricted, requiring a permit for entry.

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recommended credible worst-case Consequence would be an injury requiring medical treatment (Consequence Level 2). If a flare study has been done, then the results may justify less severe Consequences.

4.17.2 Liquid Carryover to Flare – Rain-Out

Scenario: An abnormally high flow of liquid into a flare stack (e.g., due to failure of level control or pump failure on the inlet KO vessel), leading to liquid discharge from the top of the flare. Liquid carryover from a flare can result in rain-out of ignited liquid droplets, emission of large liquid slugs that can form a pool fire, or flame-out (leading to dispersion of unburned hydrocarbons). These events can cause burn injuries to exposed persons, damage to the flare and nearby equipment, grass fires, and hydrocarbon soiling of ground and property that requires remediation.

Analysis of Consequences:

- The impact in the category of PAF – Human Injury and Illness should not be deemed credible if the following control measures are in place: (i) restricted access is provided around the flare that is controlled by a barrier such as a fence, and (ii) the entry to the restricted access area is considered a permitted activity.
- If the above control measures are not in place, then a credible worst case PAF Consequence of up to Level 2 (Medical Treatment – Lost Work Days) should be assumed based on the premise that: (i) liquid discharge from the flare should be a self-alarming event (visually and acoustically) and (ii) egress is generally not restricted such that personnel should be able to safely evacuate the hazard zone.
- For a large, slug-type flammable liquid discharge from a flare which could lead to a pool fire, a PD/LOR Consequence based on damage to the flare stack, adjacent equipment and loss of revenue due to potential shut-in production should also be evaluated. Additionally, an environmental impairment Consequence for clean-up/remediation (ECR) should be considered.
- In case of a high water-cut (>80%) liquid coming into the flare, a more likely scenario would be flare flame-out instead of ignited droplets rain-out or a pool fire.

4.17.3 Flare Flame-Out

Flare flame-out may occur due a combination of factors such as pilot failure and high winds or liquid carryover. For this scenario, there may be a potential for the flare gas to reach grade and present a flammable hazard, if ignited, with impacts to people onsite or offsite, and potential property damage. Additionally, people could be adversely affected if the gas stream contains appreciable amounts of H₂S or CO₂.

If a flare study has not been conducted prior to a PHA session, the team should consider a flame-out scenario as having at least a Consequence Level 3 for the Human Injury at a Facility category, with a follow-up recommendation to complete a full flare study. Once

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a flare study has been conducted, then the gas dispersion results may support lower Consequence levels for the applicable Consequence categories.

4.17.4 Flare Noise

Given that chronic exposure to a high noise level from a flare would be necessary for injurious effects, and that a hearing protection program is required at facilities, potential harm from this hazard can be considered non-credible. However, a high noise level can be a nuisance for public locations close to the facility and if so, this should be assigned a Level 1 Consequence for POF – People Outside a Facility. Flare noise can also be considered an in-plant nuisance and, if so, recorded as an operability issue.

4.18 Estimation of Property Damage / Loss of Revenue (PD-LOR)

The OOG Risk Matrix describes Consequence levels in this category in relatively broad ranges of financial loss. The PHA team should strive to identify the equipment potentially affected by a hazard scenario and estimate the repair or replacement cost via historical experience / knowledge to establish the property damage (PD) component of the estimate. For the loss of revenue estimate (LOR), the team should first explore for what period of time the same damaged equipment may be out of service before it could be repaired or replaced, spare systems or equipment could be started-up, or a temporary change implemented to allow operation without the affected equipment. As needed, BU Project Managers or Asset Integrity Group should be consulted to confirm a reasonable estimate of the damage and replacement costs, as well as recovery times. The loss of revenue should then be calculated using the rate of production at risk (not capacity) and the current AFE long-term product sales price estimation available from BU planning group. There should be a brief entry made in the PHA worksheet next to each scenario to include this PD-LOR estimate supporting data. If this data is not available or there is a high level of uncertainty, then the PHA team should apply their best judgment and add a recommendation to refine their estimate.

4.19 Potential Elevated Consequences (PEC) Category

Per OOG procedure 60.400.301- HSE Risk Assessment and Response, PEC codes are to be included in the PHA worksheets when a Consequence Level of 5 is assigned. PEC-1 corresponds to greater than 14 PIIIs and PEC-2 describes a potential PD-LOR of greater than \$1 billion. It should be noted that these thresholds are to be established on the basis of single distinct Event Scenario or loss event. In addition, for the PEC-1 threshold to be triggered both onsite and offsite potential PIIIs are expected to be counted together.

As an example, a central gas processing plant may handle a vast amount of associated gas from a surrounding oil production field. An equipment overpressure LOPC scenario based on blocked outlet on an inlet separator can lead to a significant escalation and damage to a large portion of the gas plant. If the plant will be shut down for several months for repair and restoration, the associated field production must be shut-in (loss of revenue). If the estimated property damage and loss of production figures exceed \$1B, PEC-2 would be triggered.

5. ABBREVIATIONS

Abbreviation	Term
CME	Critical Mitigation Element
GUL	Guided Wave Ultrasonic Inspection
IDLH	Immediately Dangerous to Life and Health
ILI	Inline Inspection
KME	Key Mitigation Element
LOPC	Loss of Primary Containment
PD-LOR	Property Damage and Loss of Revenue
PEC	Potential Elevated Consequences

6. DEFINITIONS

Term	Definition
Consequence	Characterization of the effects of a specific set of HSE-related events, occurrences or circumstances on people, the environment, property and revenue. Consequences may be described either qualitatively or quantitatively and may pose acute or chronic effects.
Distant Personnel	Distant Personnel (receptors) are considered people who are generally far enough from a Hazard such that they unlikely to be impacted by the Event Scenario.
Event Scenario	A credible and representative postulated occurrence that has the potential to result in undesired Consequences and is directly related to Company operations. Event Scenarios consist of at least two elements: 1) an Initiating Event and 2) one or more Consequences that are expressed in terms of potential impacts. Event Scenarios are analyzed for Consequence and Probability levels to form HSE Risks.



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Term	Definition
Facility	A business location or asset currently operated or maintained by OOG, or a set of such business locations or assets for which HSE matters are consolidated under a single manager, as defined by OOG. A Facility is typically a single location or geographic area (such as a plant, field, lease or right-of-way), but OOG may define a Facility to include adjacent, interconnected or integrated manufacturing or processing plants, pipeline or gathering line systems, fields or other asset locations for which HSE matters are consolidated under a single manager.
Hazard	A characteristic, activity, arrangement, event, process, or substance that has the potential to adversely affect people, the environment, property, or revenue.
Near-Field Personnel	Personnel performing routine operational and maintenance surveillance at manned facilities that are generally considered to suffer an immediate injury due to proximity to the process equipment when LOPC is identified during a PHA study such as from vessel/piping overpressure or leaks. In PHAs, the impact to the Near-Field Personnel is the primary focus.
Permanent Incapacitating Injury	A fatality or physical injury or illness that permanently renders a person incapable of performing the essential duties of any of the occupations for which a person is qualified by education, training, and experience.
Permanent Occupied Structure	Residences (including anchored mobile home residences) with any number of occupants. Any single office, commercial or industrial establishment that is not associated with OOG operations and is likely to have >100 occupants at a given time. (For several office, commercial and industrial establishments in close vicinity <u>that may individually have less than 100 occupants</u> , aggregate occupancy within any 90° section of a circle in any direction from a release point - if estimated occupancy is >100, count as 1 POS.)
Probability	The likelihood of a given Event Scenario occurring. It may be determined with reference to historical data, analysis of protective measures, failure modeling and professional judgment.
Process Hazard Analysis (PHA)	A systematic approach for identifying and assessing Process hazards and risks and their possible effects





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7. REFERENCES

Document	Title
API 12D	Specification for Field Welded Tanks for Storage of Production Liquids
API 12F	Specification for Shop Welded Tanks for Storage of Production Liquids
API 610	Centrifugal Pumps for Petroleum, Petrochemical and Natural Gas Industries
API 650	Welded Tanks for Oil Storage
API 682	Pumps—Shaft Sealing Systems for Centrifugal and Rotary Pumps
API Publication 937	Evaluation of Design Criteria for Storage Tanks With Frangible Roof Joints
ASME B31.3	Process Piping
ASME B31.4	Pipeline Transportation Systems for Liquids and Slurries
ASME B31.8	Gas Transmission and Distribution Piping Systems
ASME Section VIII	Boiler and Pressure Vessel Code
BS EN 12285	Workshop fabricated steel tanks
BS EN 14015	Specification for the design and manufacture of site built, vertical, cylindrical, flat-bottomed, above ground, welded, steel tanks for the storage of liquids at ambient temperature and above
OOG 60.400.301	HSE Risk Assessment and Response Procedure
OOG 60.400.301.01	Assigning H ₂ S Consequence Levels Guideline
OOG 60.400.301.03	Development of Event Scenarios
OOG 60.400.301.10	Credible Consequences & CME Credits for Liquid Carryover Scenario to a Compressor
OOG 60.400.308.01	Facility Siting and Layout Study Guideline
OOG 60.400.310	Procedure for Management of Hazards Associated with Location of Portable Buildings

